SEQUENCE 14

Probability: conditioning and weighted trees

26 April–24 May 2027

14.1 Sample space, outcomes and events

Key point · Vocabulary. A random experiment has several possible results, without it being known in advance which will occur. An outcome is a possible result; the sample space Ω is the set of them. An event is a subset of Ω; an elementary event contains a single outcome. The impossible event is $\varnothing$; the certain event is Ω.

Key point · Operations on events. $\overline{A}$ is the complementary event of A. $A \cap B$ means "A and B"; $A \cup B$, "A or B", possibly both. Two events are mutually exclusive if $A \cap B = \varnothing$; otherwise, they are compatible.

Example 1. A six-sided die is rolled. Write Ω, A: "get an even number", B: "get 4", then $\overline{A}$, $A \cap B$ and $A \cup B$. Are A, B compatible?

Solution. $\Omega = \{1;2;3;4;5;6\}$, $A = \{2;4;6\}$, $B = \{4\}$, $\overline{A} = \{1;3;5\}$, $A \cap B = \{4\}$, $A \cup B = A$. The events are compatible because they share the outcome 4.

Example 2. With $A = \{a;b;c;d;e\}$ and $B = \{b;e;f;g;h\}$, give the intersection and the union. Can $\overline{A}$ be determined without specifying the sample space?

Solution. $A \cap B = \{b;e\}$ and $A \cup B = \{a;b;c;d;e;f;g;h\}$. The complement depends on the sample space; it cannot be determined without this information.

14.2 Probability model and equally likely outcomes

Key point · Probability on a finite sample space. To each outcome, a model assigns a probability that is positive or zero; the sum is 1. The probability of an event is the sum of the probabilities of its outcomes. We have

$$P(\Omega) = 1, \quad P(\varnothing) = 0, \quad P(\overline{A}) = 1 - P(A).$$

With equally likely outcomes, all outcomes have the same probability and

$$P(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)}.$$

Equal likelihood is an assumption of the model, not a consequence of the list of outcomes alone.

Example 3. Assume the die is fair. For A = "even" and C = "greater than or equal to 5", calculate $P(A)$, $P(C)$, $P(A \cap C)$ and $P(A \cup C)$.

Solution. $P(A) = 3/6 = 1/2$, $P(C) = 2/6 = 1/3$, $A \cap C = \{6\}$ so $P(A \cap C) = 1/6$. $A \cup C = \{2;4;5;6\}$, so $P(A \cup C) = 4/6 = 2/3$. The outcome 6 must not be counted twice.

Example 4. A drawing pin can land point up or point down. Can we deduce that each outcome has probability $1/2$? How can a model be proposed?

Solution. No: two outcomes are not enough to impose equal likelihood. One can repeat the throws under comparable conditions and use the observed frequencies to propose a model, bearing in mind fluctuations and the characteristics of the object.

Teacher note. Reactivate the link between observed frequency and postulated probability. An experimental result does not prove exact equal likelihood.

14.3 Conditional probability from a table

Key point · Definition. For $P(A) > 0$, the probability of B given A is

$$P_A(B) = \frac{P(A \cap B)}{P(A)}, \qquad P(A \cap B) = P(A) P_A(B).$$

For an equally likely draw within a finite population:

$$P_A(B) = \frac{\text{Card}(A \cap B)}{\text{Card}(A)}.$$

We also write $P(B \mid A)$. $P_A(B)$ and $P_B(A)$ are generally not equal.

Example 5. A student is drawn uniformly from the class of 32 students studied earlier. G: "boy", N: "takes CS".

	G	$\overline{G}$	Total
N	12	9	21
$\overline{N}$	6	5	11
Total	18	14	32

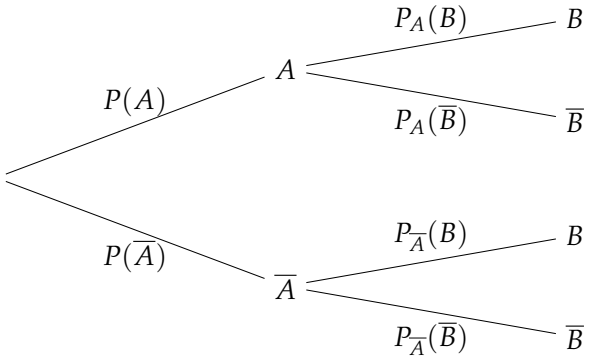
Calculate $P(G)$, $P(N)$, $P(G \cap N)$, $P_N(G)$ and $P_G(N)$.

Solution. $P(G) = 18/32 = 9/16$, $P(N) = 21/32$, $P(G \cap N) = 12/32 = 3/8$. $P_N(G) = 12/21 = 4/7$, whereas $P_G(N) = 12/18 = 2/3$. By the formula: $P_N(G) = (12/32)/(21/32) = 12/21$. Given that the student takes CS, the reference population is reduced to the 21 students concerned.

Watch out. Conditioning requires the known event to have nonzero probability; we do not divide by zero.

14.4 Weighted trees and the probability of a path

Key point · Rules. At each node, the sum of the probabilities of the outgoing branches is 1. A second-level branch carries a conditional probability. A path represents an intersection of events; its probability is the product of the probabilities written along that path.



Example 6. We know $P(A) = 0.6$, $P_A(B) = 0.7$ and $P_{\overline{A}}(\overline{B}) = 0.9$. Complete the other branches and calculate the probabilities of the four paths.

Solution. $P(\overline{A}) = 0.4$, $P_A(\overline{B}) = 0.3$, $P_{\overline{A}}(B) = 0.1$.

$$P(A \cap B) = 0.6 \times 0.7 = 0.42, \quad P(A \cap \overline{B}) = 0.18,$$

$$P(\overline{A} \cap B) = 0.04, \qquad P(\overline{A} \cap \overline{B}) = 0.36.$$

The sum of the four probabilities is 1, which checks the consistency of the model.

Example 7. Justify the multiplication along the path A then B from the definition of $P_A(B)$.

Solution. Since $P_A(B) = P(A \cap B)/P(A)$ and $P(A) > 0$, multiplying by $P(A)$ gives $P(A)P_A(B) = P(A \cap B)$.

14.5 From a table to a tree

Example 8. For the class of 32 students, construct a tree starting with CS / no CS, then boy / girl. Then construct the tree in the reverse order. Check, in both trees, the probability of choosing a girl who takes CS.

Solution. First tree: initial branches $P(N) = 21/32$, $P(\overline{N}) = 11/32$. After N: $P_N(G) = 12/21 = 4/7$, $P_N(\overline{G}) = 9/21 = 3/7$. After $\overline{N}$: $6/11$ and $5/11$. Second tree: initial branches $P(G) = 18/32 = 9/16$, $P(\overline{G}) = 14/32 = 7/16$. After G: $P_G(N) = 12/18 = 2/3$, then $1/3$. After $\overline{G}$: $P_{\overline{G}}(N) = 9/14$, then $5/14$. The required path is worth $(21/32)(9/21) = 9/32$ in the first tree and $(14/32)(9/14) = 9/32$ in the second. Changing the order preserves the probabilities of the intersections but changes the probabilities of the branches.

Example 9. An urn contains 3 red balls and 2 blue balls. Two balls are drawn in succession **without replacement**. Construct the tree of colours and calculate the probability of drawing two red balls.

Solution. First level: R with $3/5$, B with $2/5$. After R, 2 red and 2 blue remain: branches $1/2, 1/2$. After B, 3 red and 1 blue remain: branches $3/4, 1/4$. So $P(R_1 \cap R_2) = (3/5)(1/2) = 3/10$. Using $3/5$ on the second red branch would ignore the absence of replacement.

14.6 Inverting conditioning and false positives

Example 10 · Fictional component check. A batch contains 10000 components, of which 100 are defective. A test is positive for 99 of the defective components and for 495 of the non-defective components. A component is chosen uniformly. D: "defective"; T: "test positive".

1. Complete the two-way table of test / true state.
2. Calculate $P_D(T)$ and $P_T(D)$.
3. Give the number of false positives and false negatives.
4. Calculate the sensitivity $P_D(T)$ and the specificity $P_{\overline{D}}(\overline{T})$.

Solution.

	D	$\overline{D}$	Total
T	99	495	594
$\overline{T}$	1	9405	9406
Total	100	9900	10000

$P_D(T) = 99/100 = 99\,\%$. On the other hand $P_T(D) = 99/594 = 1/6 \approx 16.67\,\%$. A positive test therefore does not imply here a 99 % probability of defect. The 495 false positives are the sound components that tested positive; the false negative is the defective component that tested negative. The sensitivity is 99 %; the specificity $9405/9900 = 95\,\%$. The low initial proportion of defective components explains why false positives outnumber true positives despite the good sensitivity.

Key point · Reading a test result. "Positive among the defective" and "defective among the positives" correspond to two different conditionings. Always name the reference group before the calculation.

Teacher note. The numbers are fictional. The table allows studying the inversion without requiring a general total-probability formula: this is not an expectation of the 2026 Seconde programme.

14.7 Simulation and the law of large numbers

Key point · Popularised version. When an experiment is repeated a large number of times in the same model with independent repetitions, the observed frequency of an event is, barring exceptions, close to its probability. It fluctuates: it does not necessarily get closer with each new trial.

```
from random import Random

def frequences_six(n, graine=1):
    # n entier >= 1 ; modele de lancers independants
    rng = Random(graine)
    succes = 0
    frequences = []
    for k in range(1, n+1):
        if rng.randint(1, 6) == 6:
            succes = succes+1
        frequences.append(succes/k)
    return frequences
```

Example 11. Plot the frequencies for $n = 1000$, with several seeds. Compare with $1/6$. Why fix the seed? Can we require the final frequency to be exactly $1/6$?

Solution. The plots depend on the seed and fluctuate around $1/6$. Fixing the seed makes the simulation reproducible. With 1000 throws, a final frequency of exactly $1/6$ is even impossible, because $1000/6$ is not an integer. A nearby frequency is consistent with the model, but is not a proof of it.

Example 12. Modify the program to track the frequency of the event "even result". Which reference probability should be used?

Solution. Replace the test with `rng.randint(1, 6) % 2 == 0`. The reference probability is $3/6 = 1/2$. Each trial must make a single draw; do not call the generator several times to test the same result.